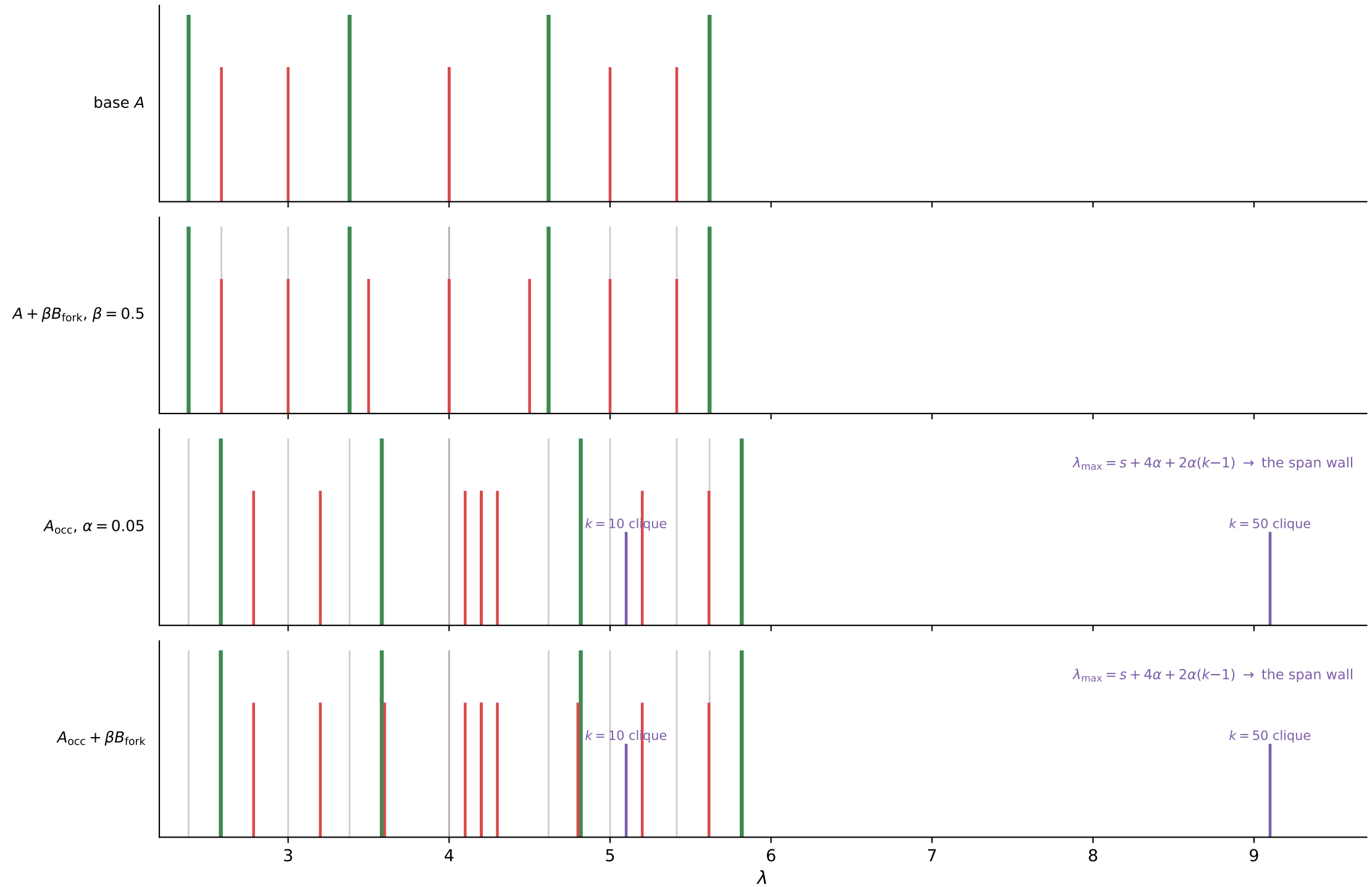


The motif lines under each operator (exact closed forms; grey = base-A positions)



green: true-track P_4 lines $s+4\alpha-2\cos(k\pi/5)$ · red: false motifs — isolated $s+4\alpha$, chains $m = 2, 3$, competing pair $s+4\alpha \mp (2\alpha+\beta)$, same-role pair $s+2\alpha, s+6\alpha$ · purple: same-role k -clique top line (occupancy span mechanism) · $s = 4, \alpha = 0.05, \beta = 0.5$